



STM Exercise 9

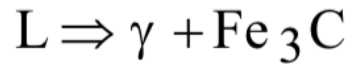
Phase Diagrams (Fe-C)

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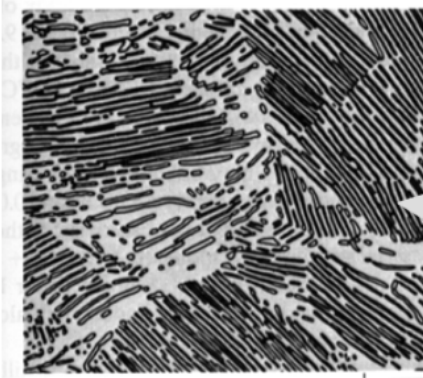
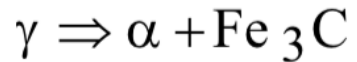
IRON-CARBON (Fe-C) PHASE DIAGRAM

- 2 important points

-Eutectic (A):

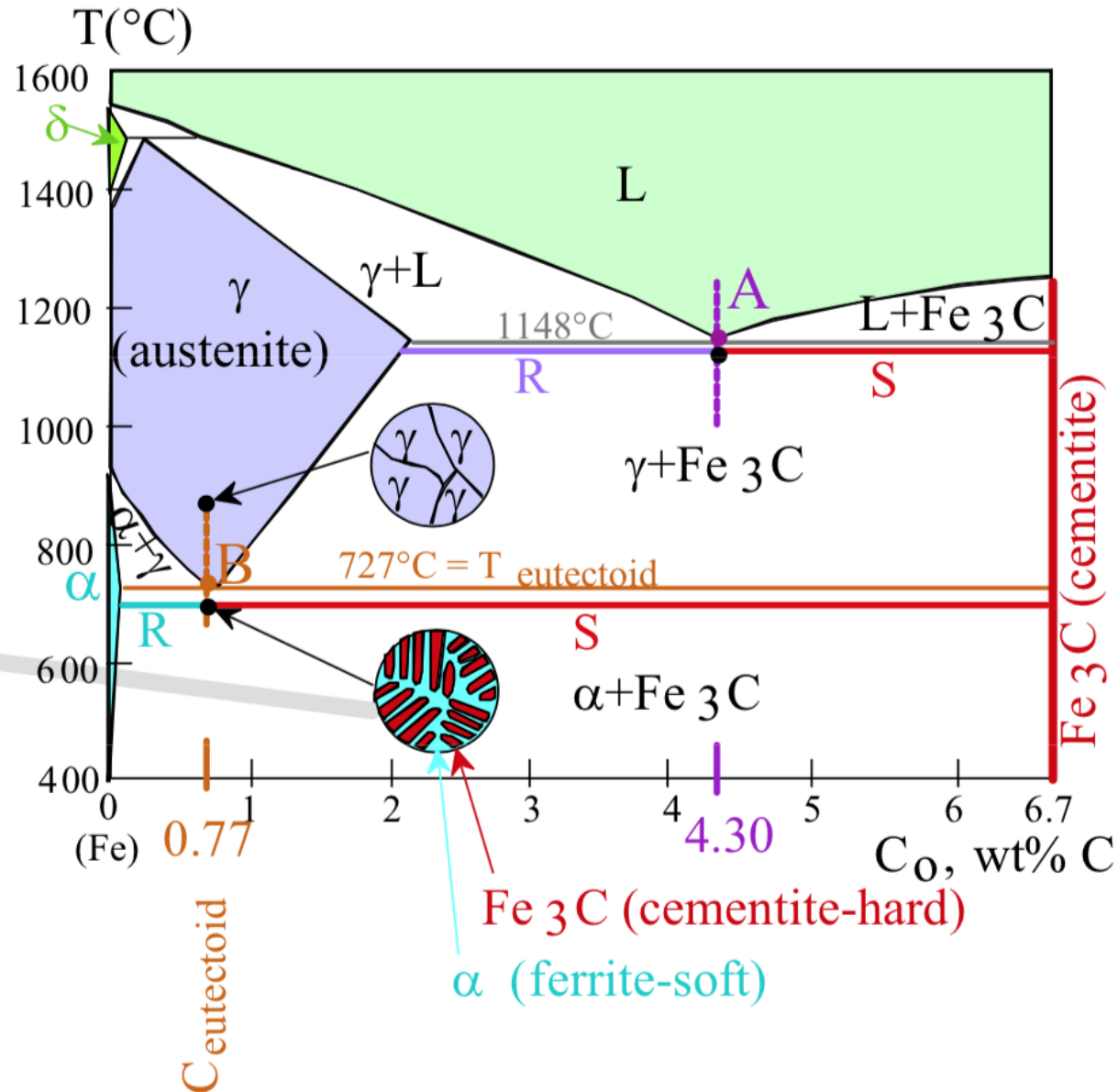


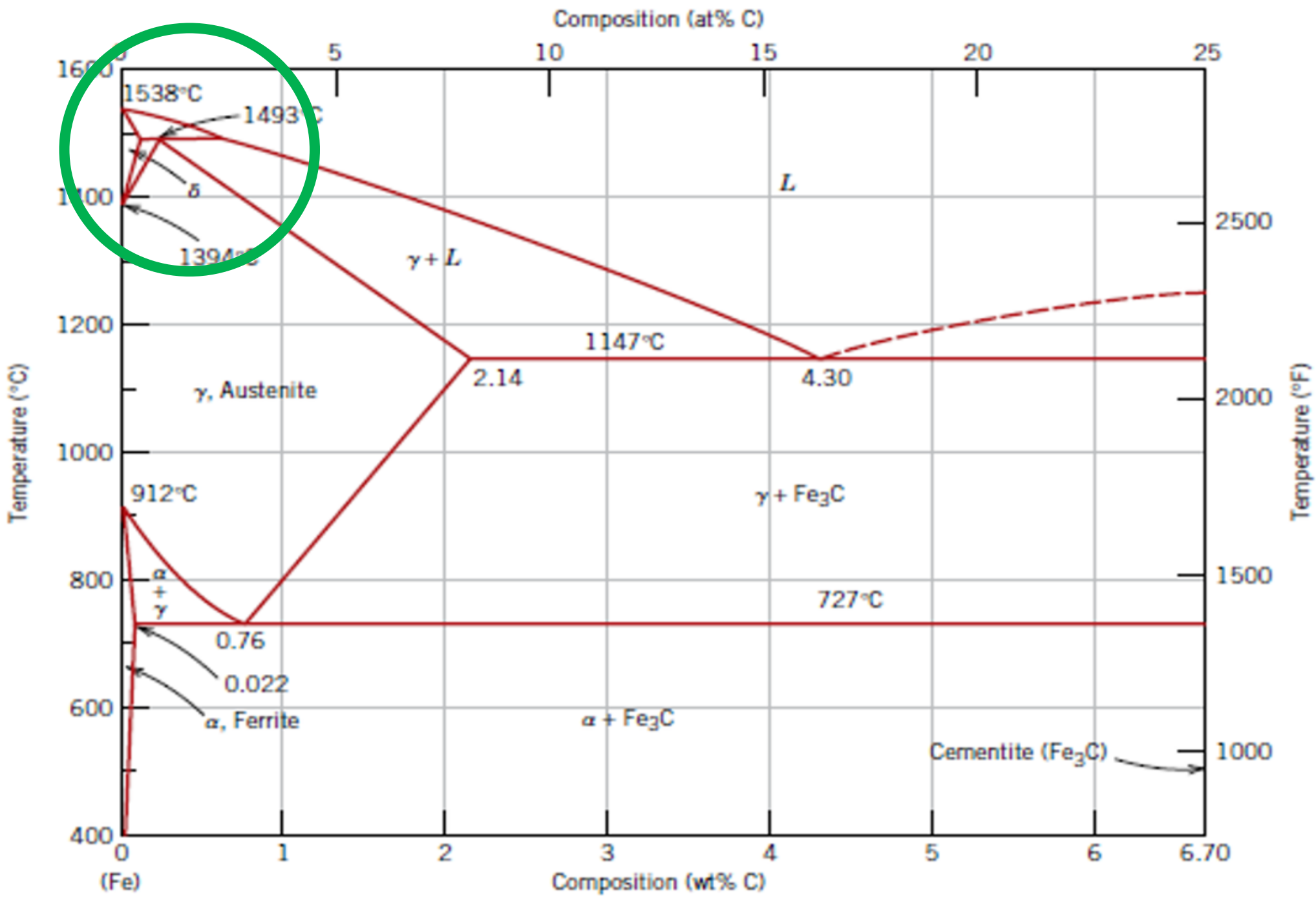
-Eutectoid (B):

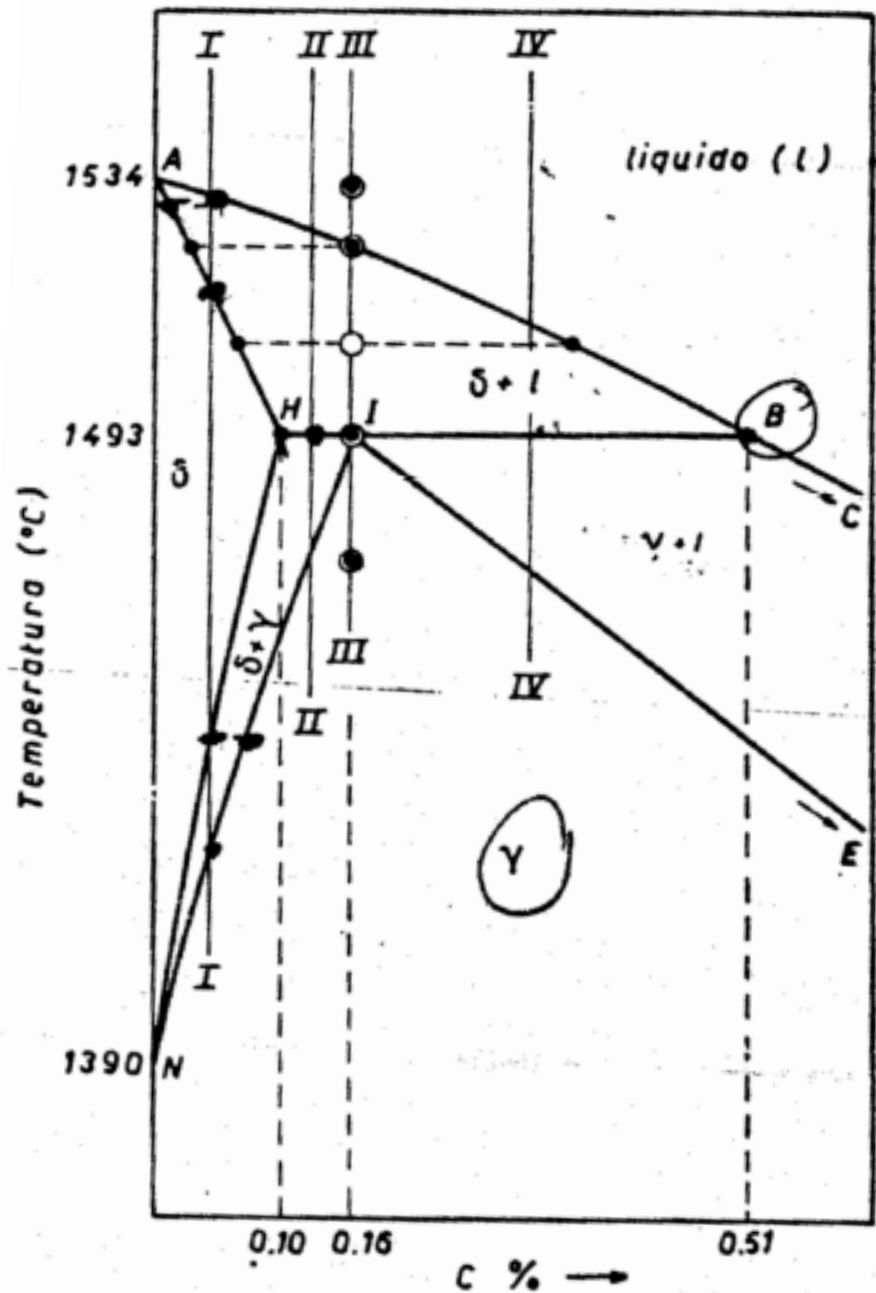


120 μm

Result: Pearlite = alternating layers of α and Fe_3C phases.

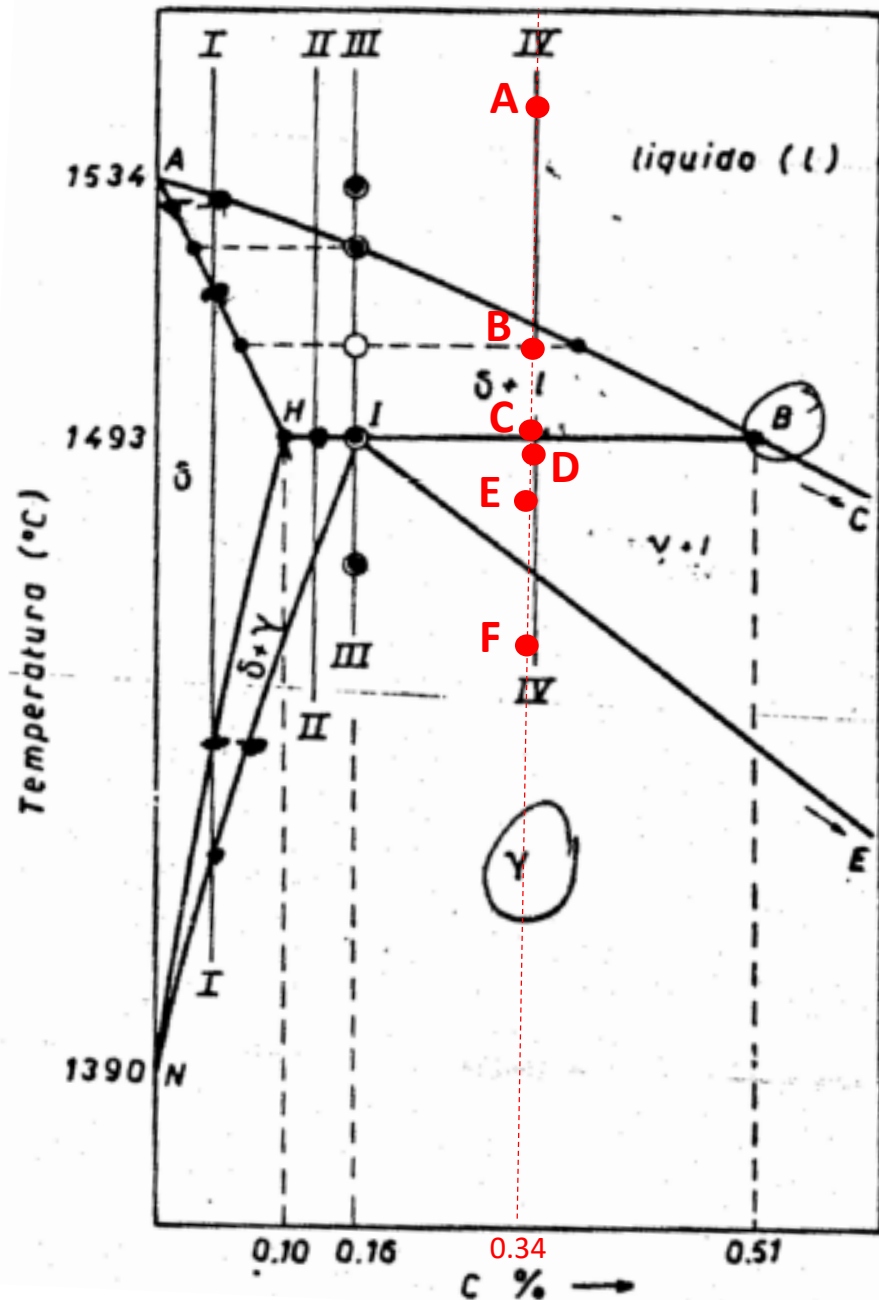




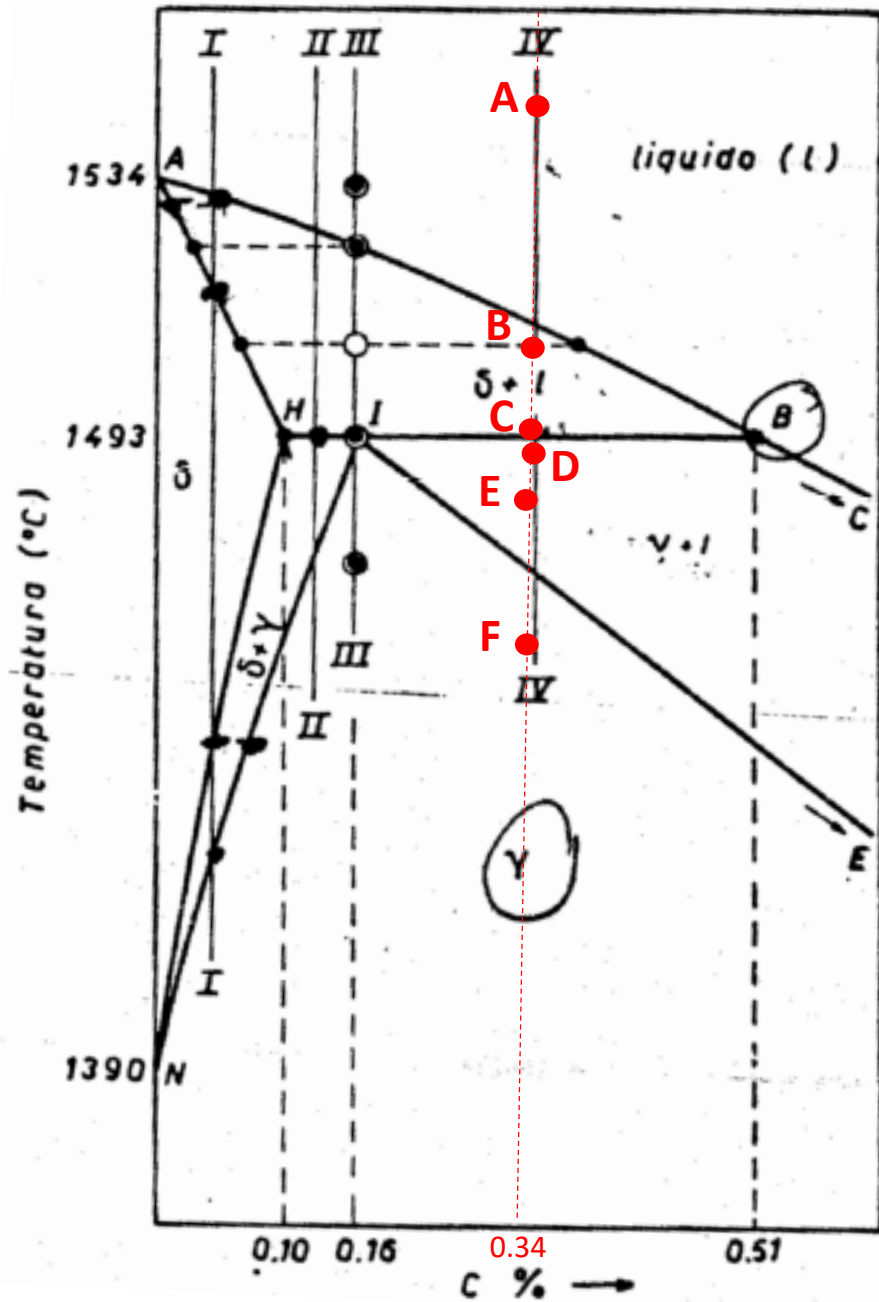


Exercise 1: which transformation occurs in point I?

Point I is a **PERITECTIC** transformation
 $\delta + L \leftrightarrow \gamma$



Exercise 2: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), E and F for a steel with 0.34%C.

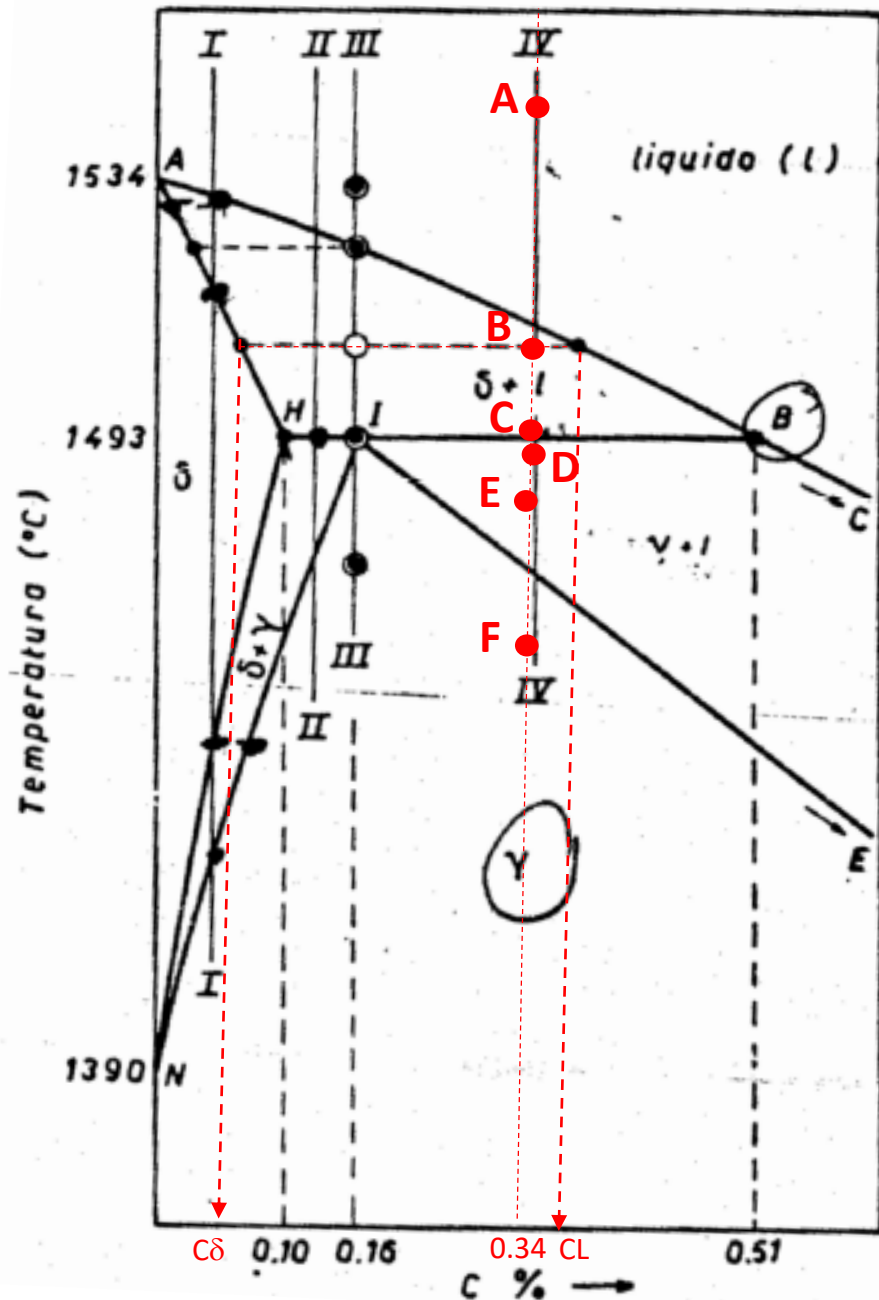


Point A:

1 phase liquid

Composition 0.34%C

Relative amount 100% liquid



Point B:

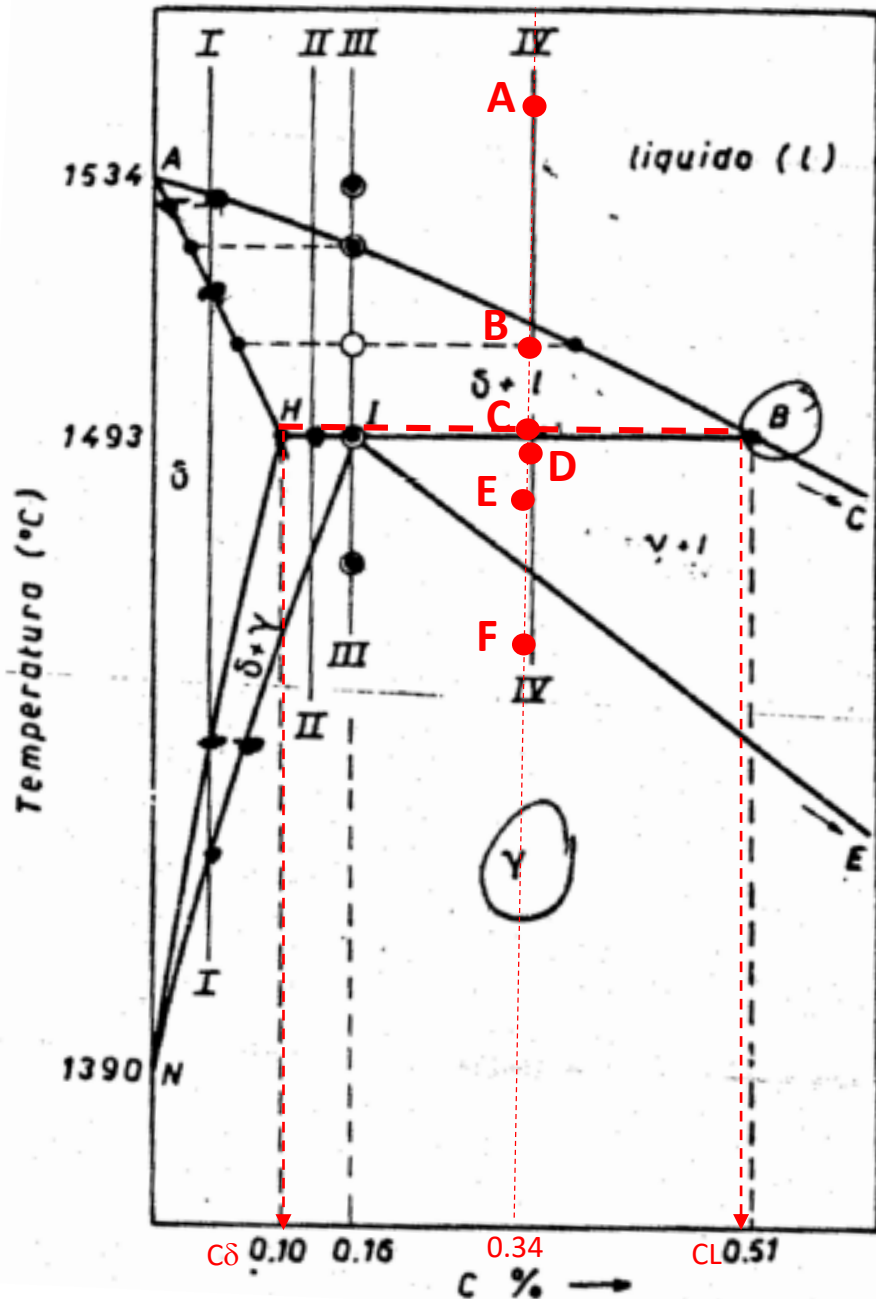
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.05\%C$
- Composition of liquid: $CL=0.4\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.4-0.34)/(0.4-0.05)*100=17\%$
- $\% \text{Liquid} = (0.34-0.05)/(0.4-0.05)*100=83\%$



Point C:

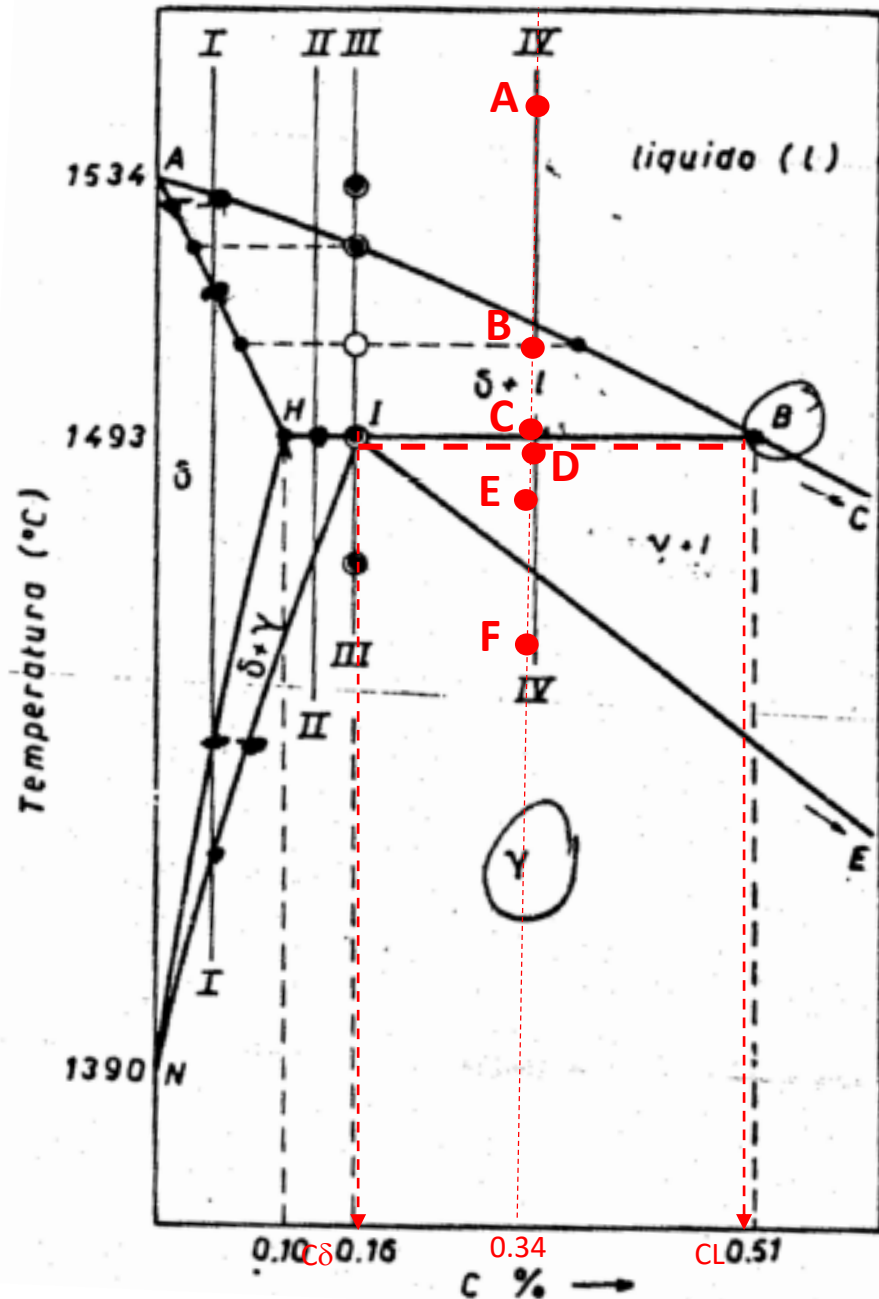
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.1\%C$
- Composition of liquid: $CL=0.51\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.51-0.34)/(0.51-0.1)*100=41.5\%$
- $\% \text{Liquid} = (0.34-0.1)/(0.51-0.1)*100=58.5\%$



Point D:

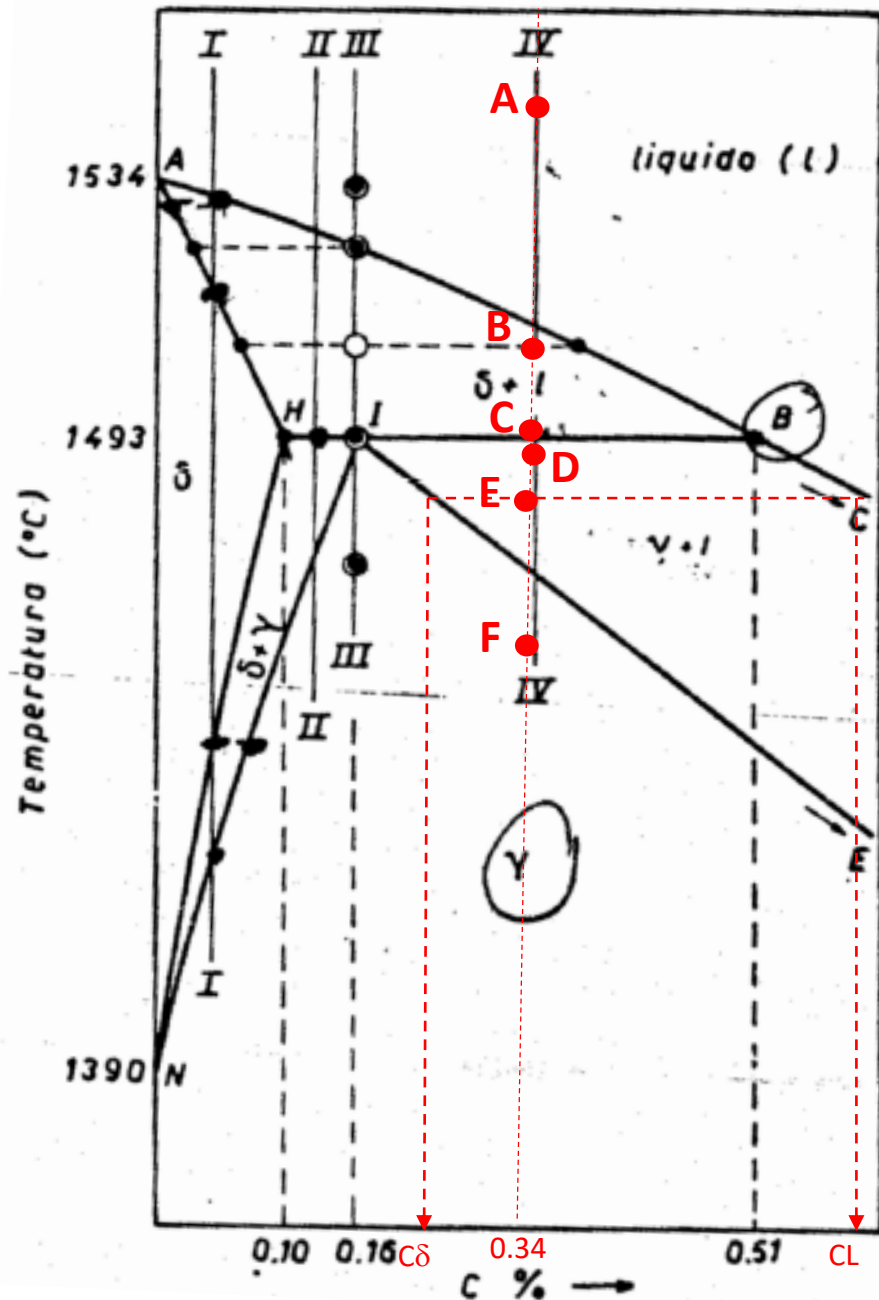
2 phases γ austenite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of γ austenite : $C_{\delta}=0.16\%C$
- Composition of liquid: $CL=0.51\%C$

Relative amounts (LEVER RULE)

- $\% \gamma \text{ austenite} = (0.51-0.34)/(0.51-0.16)*100=48.5\%$
- $\% \text{Liquid} = (0.34-0.16)/(0.51-0.16)*100=51.5\%$



Point E:

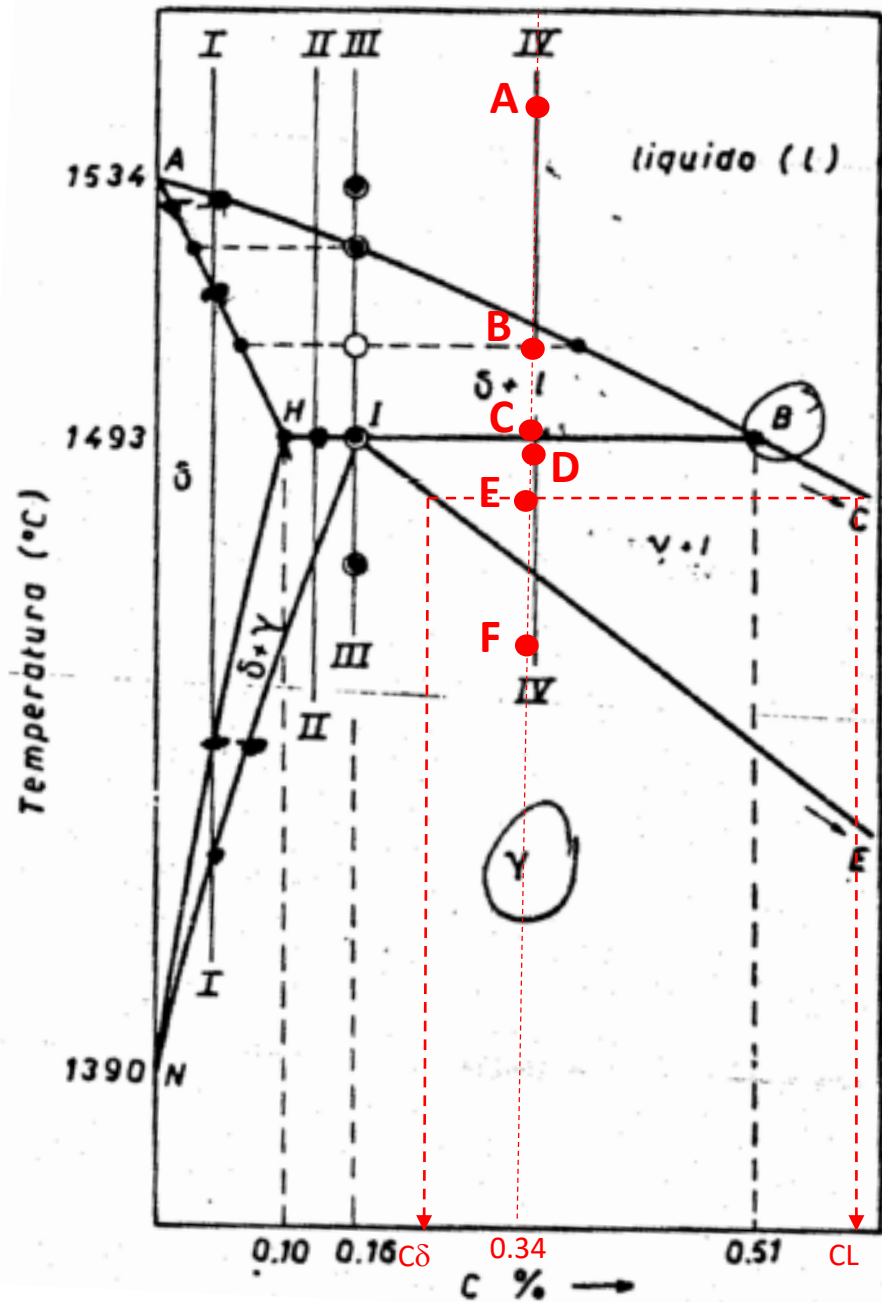
2 phases γ austenite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of γ austenite : $C_{\delta}=0.2\%C$
- Composition of liquid: $CL=0.6\%C$

Relative amounts (LEVER RULE)

- $\% \gamma \text{ austenite} = (0.6-0.34)/(0.6-0.2)*100=65\%$
- $\% \text{Liquid} = (0.34-0.2)/(0.6-0.2)*100=35\%$

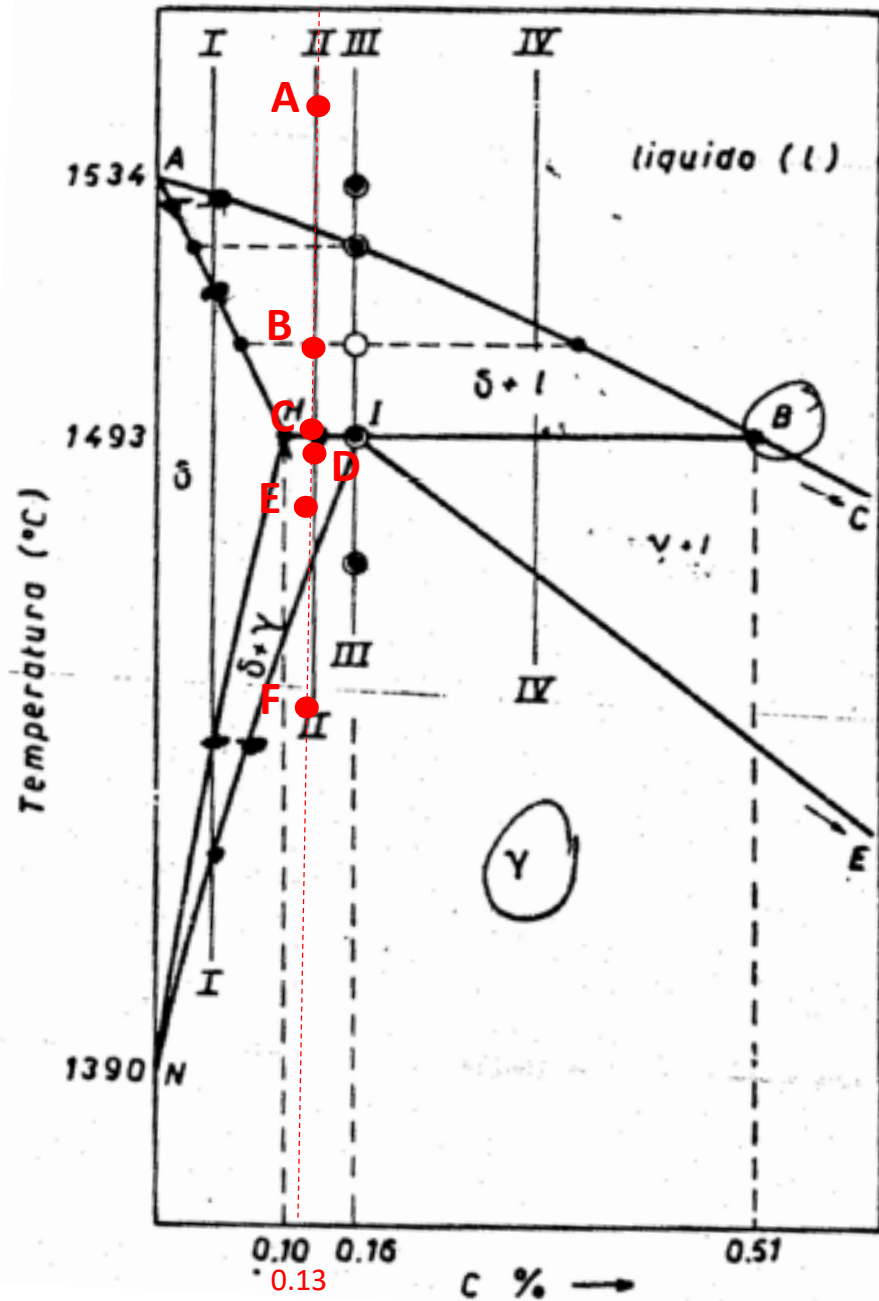


Point F:

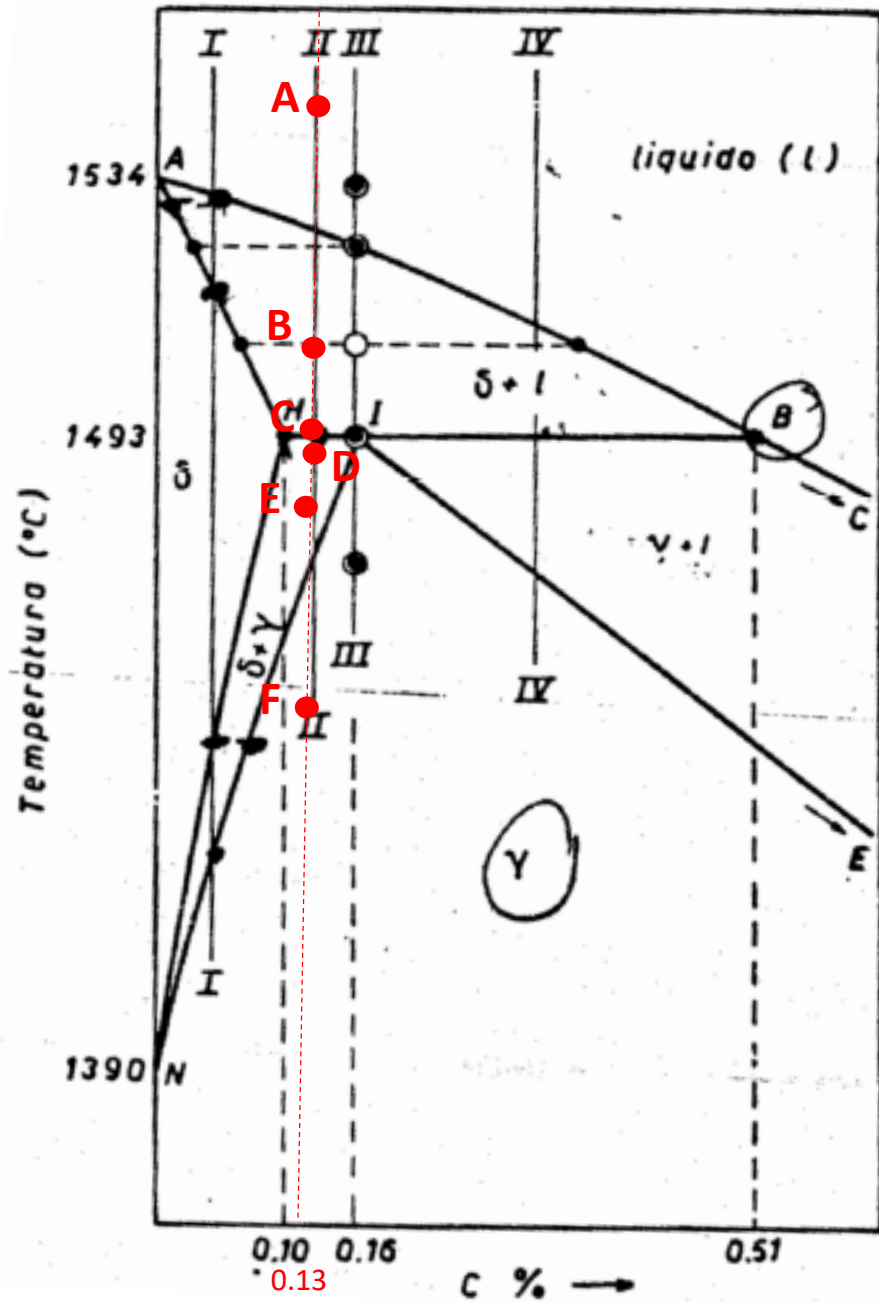
1 phase γ austenite (solid)

Composition: 0.34%C

Relative amount 100% solid γ austenite



Exercise 3: Determine the phase/s present their compositions and relative amount in points A, B, C (at the beginning of the peritectic transformation), D(at the end of the peritectic transformation), E and F for a steel with 0.13%C.

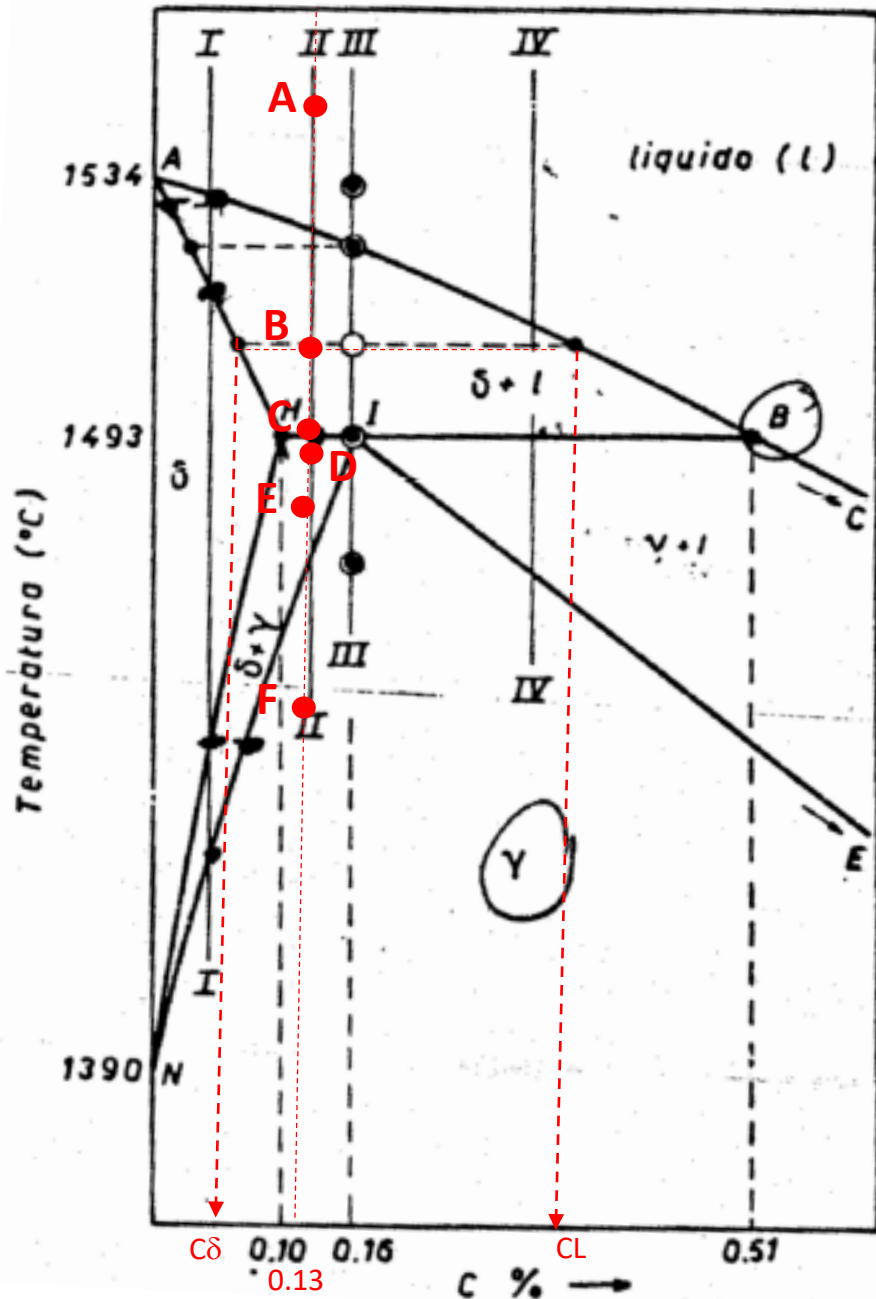


Point A:

1 phase liquid

Composition 0.13%C

Relative amount 100% liquid



Point B:

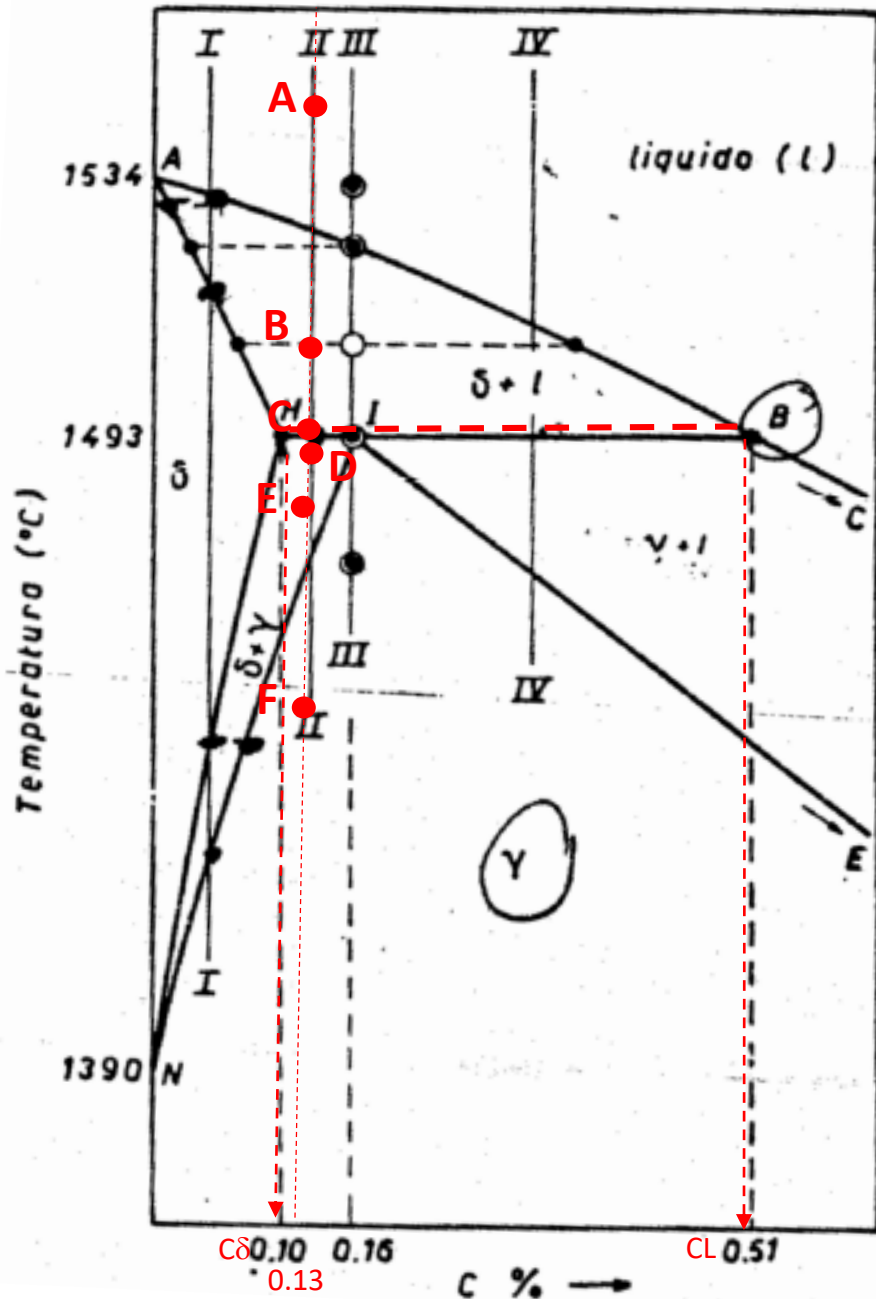
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.05\%C$
- Composition of liquid: $CL=0.4\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.4-0.13)/(0.4-0.05)*100=77\%$
- $\% \text{Liquid} = (0.13-0.05)/(0.4-0.05)*100=23\%$



Point C:

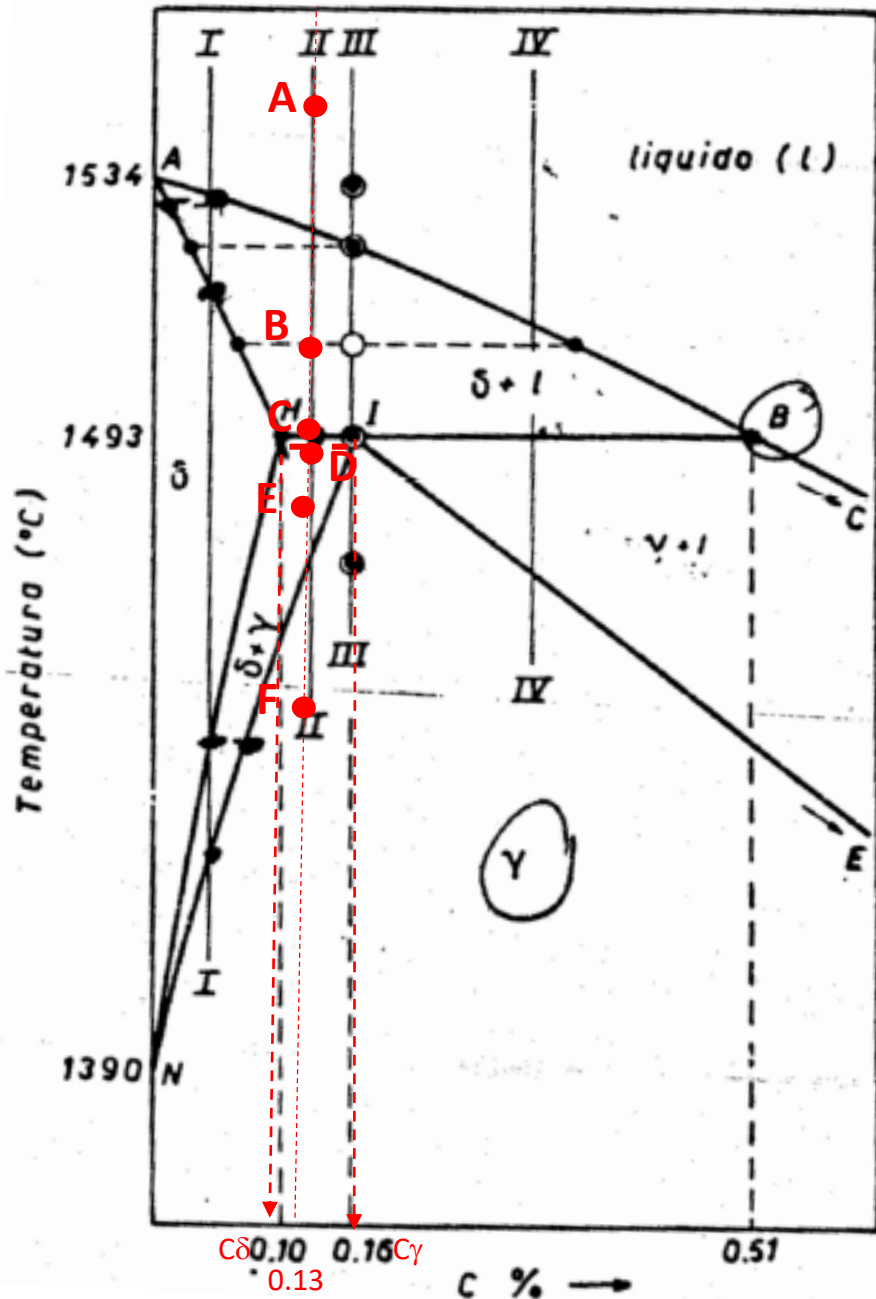
2 phases δ ferrite (solid) and liquid

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.1\%C$
- Composition of liquid: $CL=0.51\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.51-0.13)/(0.51-0.1)*100=93\%$
- $\% \text{Liquid} = (0.13-0.1)/(0.51-0.1)*100=7\%$



Point D:

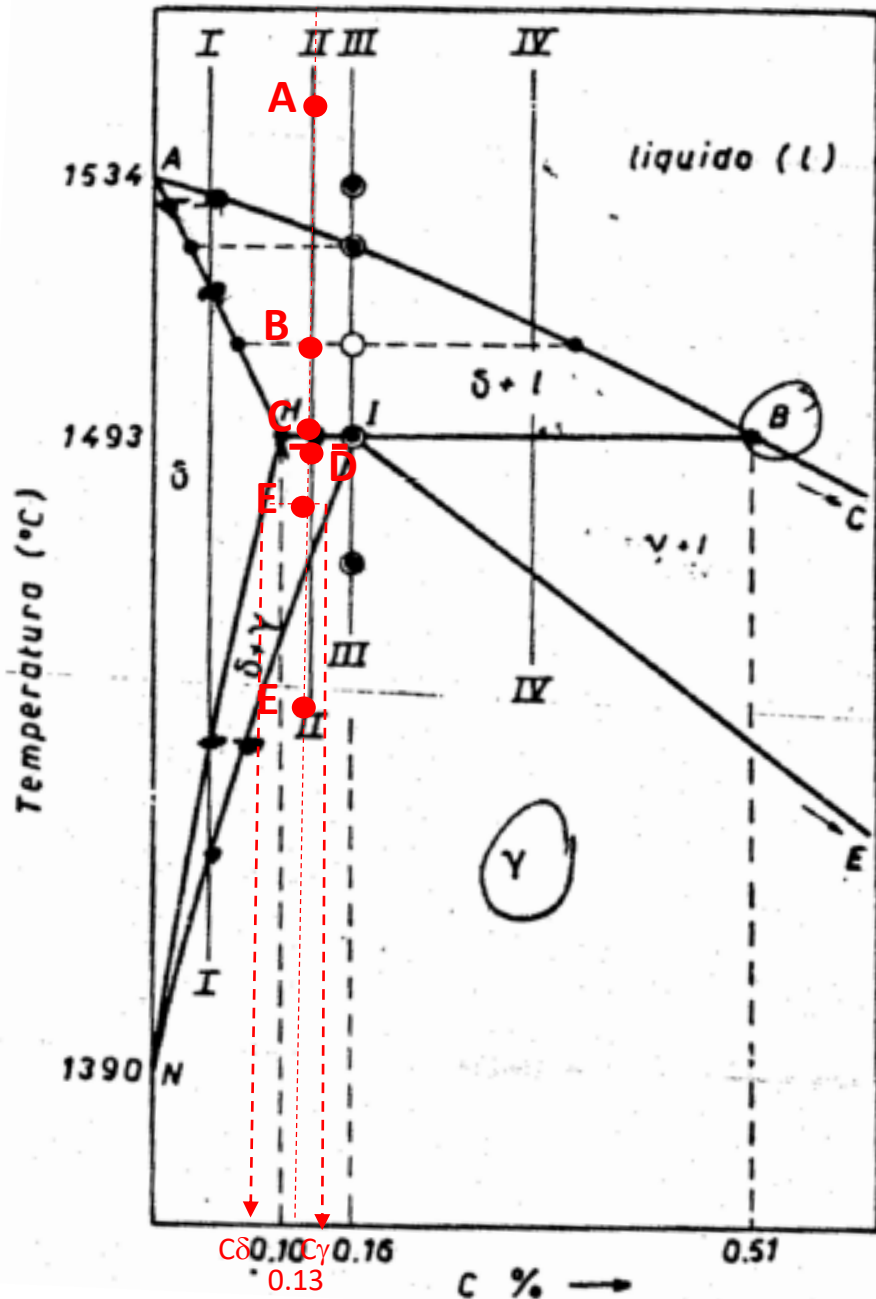
2 phases δ ferrite (solid) and γ austenite (solid)

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.1\%C$
- Composition of γ austenite (solid): $C_{\gamma}=0.16\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.16-0.13)/(0.16-0.1)*100=50\%$
- $\% \gamma \text{ austenite} = (0.13-0.1)/(0.16-0.1)*100=50\%$



Point E:

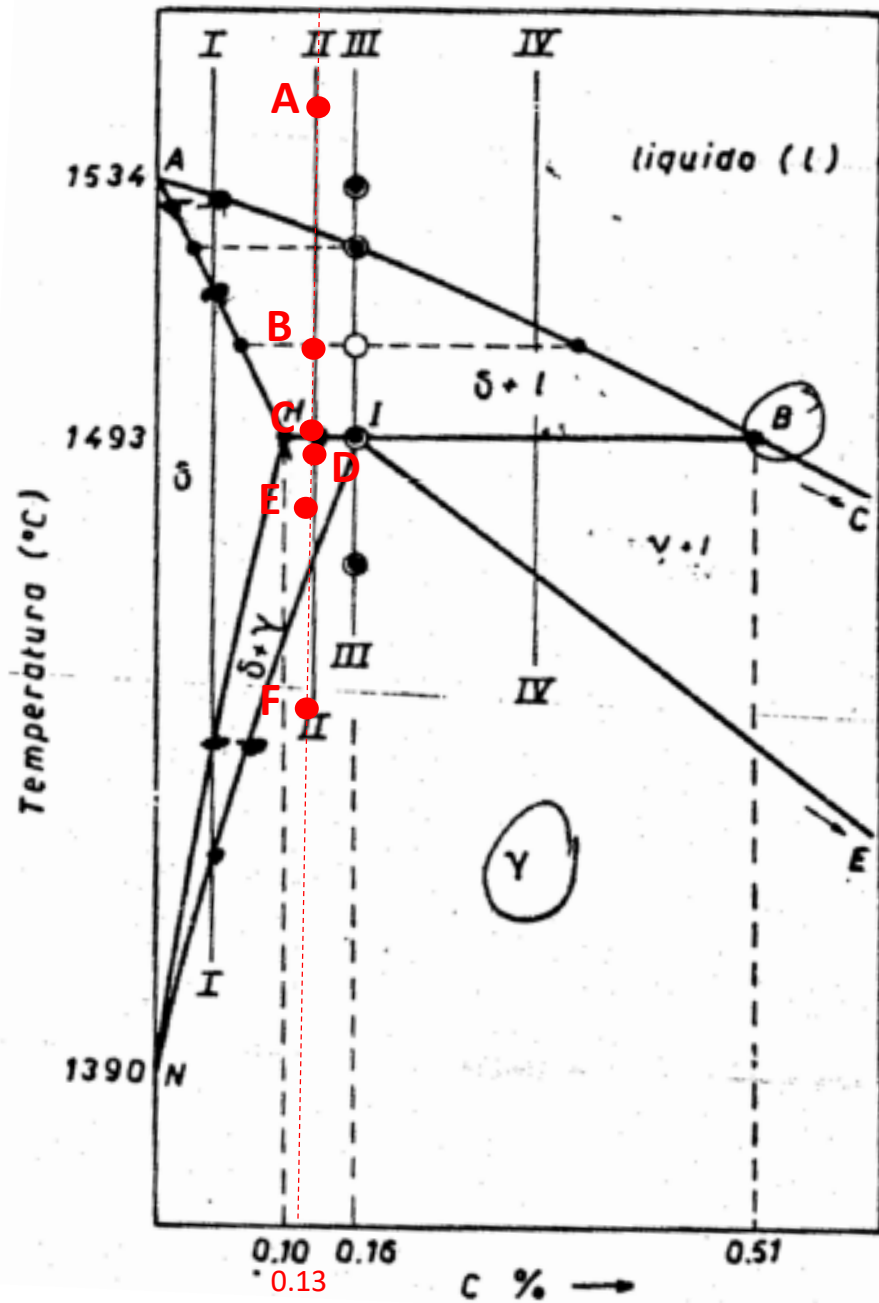
2 phases δ ferrite (solid) and γ austenite (solid)

Compositions (HORIZONTAL RULE)

- Composition of δ ferrite : $C_{\delta}=0.08\%C$
- Composition of γ austenite (solid): $C_{\gamma}=0.14\%C$

Relative amounts (LEVER RULE)

- $\% \delta \text{ ferrite} = (0.14-0.13)/(0.14-0.08)*100=17\%$
- $\% \gamma \text{ austenite} = (0.13-0.08)/(0.14-0.08)*100=83\%$



Point F:

1 phase γ austenite (solid)

Composition: 0.13%C

Relative amount 100% solid γ austenite

Questions on exercises?

